

Engineering Tolerance Tables

Engineering · 2025-04 · Revision 9

1. General tolerances

General tolerances apply where no tolerance is stated against a dimension on the drawing. Where a tolerance is stated on the drawing, the drawing governs.

Nominal range	Fine	Medium	Coarse
0.5 to 3	0.05	0.10	0.20
over 3 to 6	0.05	0.10	0.30
over 6 to 30	0.10	0.20	0.50
over 30 to 120	0.15	0.30	0.80
over 120 to 400	0.20	0.50	1.20

Tolerance class is stated in the drawing title block. Where no class is stated, medium applies.

2. Machined faces

Form tolerances for machined faces are stated by feature and by tolerance class. Values are in millimetres and apply over the full extent of the feature unless a partial extent is stated on the drawing.

Feature	Class A	Class B	Class C
Valve body, port face	0.02	0.04	0.08
Valve body, mounting pad	0.03	0.06	0.12
Valve body, bore land	0.02	0.03	0.06
Valve body, seat register	0.01	0.02	0.05
Valve body, end face	0.04	0.08	0.15
Cover plate, outer face	0.05	0.10	0.20
Cover plate, inner face	0.03	0.06	0.12
Cover plate, spigot	0.02	0.04	0.08
Manifold block, top face	0.04	0.08	0.16
Manifold block, side face	0.06	0.12	0.24
Manifold block, port register	0.02	0.04	0.08
Adapter, flange face	0.03	0.06	0.12
Adapter, spigot register	0.02	0.04	0.08
Retainer, seating face	0.02	0.05	0.10
Retainer, outer diameter	0.05	0.10	0.20
Spacer, both faces	0.04	0.08	0.16
Gland, bore	0.02	0.04	0.08
Gland, face	0.03	0.06	0.12
BV-12 housing, flange face	0.03	0.06	0.12
BV-12 housing face, flatness	0.02	0.05	0.10
BV-13 housing face, flatness	0.03	0.07	0.14
BV-12 housing, bolt circle	0.05	0.10	0.20

Flatness is measured over the full face with the component unclamped and at ambient temperature. Measurement with the component held in the assembly fixture is not equivalent and shall not be recorded against this table.

Class is assigned per feature in the drawing title block. Where a feature appears in this table and is not assigned a class on the drawing, Class B applies.

3. Fastener torque bands

Torque bands state the acceptable range around a specified value. The band is expressed as a percentage of the specified value and applies at the point of application.

Fastener	Specified value basis	Band, revision 8	Band, revision 9
M6	Per instruction	±20%	±12%
M8	Per instruction	±20%	±10%
M10	Per instruction	±20%	±10%
M12	Per instruction	±20%	±12%

Bands stated at revision 8 applied a single tolerance across all fastener dimensions. Revision 9 states a band per dimension.

Provision	Revision 8	Revision 9
Band basis	Single, all dimensions	Per dimension
Measurement point	Not stated	At application
Residual measurement	Not covered	Covered, see note

Residual torque measured after a settling interval is not within the scope of the band stated at application. Where a residual measurement is required, the acceptance value is stated in the instruction governing the assembly.